MILESTONE-4

Project Title: ExoHabitAI – Mapping the Goldilocks Zone

1. Introduction

ExoHabitAI is an AI-powered web application designed to analyze exoplanet habitability using astrophysical data from NASA's Kepler Mission. The system leverages machine learning techniques to classify planets as habitable or non-habitable based on multiple stellar and planetary parameters.

The application provides an interactive dashboard where users can explore datasets, run predictions, and visualize model performance.

2. Problem Statement

Identifying habitable exoplanets is a complex challenge due to:

- High-dimensional astrophysical data
- Noisy observations from telescopes
- Lack of direct measurement of life-supporting conditions

This project aims to simplify the process by:

- Using machine learning to predict habitability
- Providing a user-friendly interface for analysis
- Automating feature extraction and prediction

3. Objectives

- To analyze Kepler dataset with 40+ astrophysical features
- To build a machine learning model for habitability prediction
- To develop an interactive web-based dashboard
- To allow both single and batch prediction of planets
- To visualize model performance and insights

4. System Architecture

Frontend

- Built using modern UI design (HTML, CSS, JavaScript)
- Interactive dashboard with multiple sections:
  - Launchpad
  - Mission Overview
  - Predictor
  - Batch Processing
  - Metrics

Backend

- Developed using Flask API
- Handles:
  - Data preprocessing
  - Feature alignment
  - Model inference

Machine Learning Model

- Ensemble model (500 estimators)
- Uses:
  - Feature importance analysis
  - Bootstrapping for robustness
- Outputs:
  - Prediction (Habitable / Non-Habitable)
  - Probability score

5. Working of the System

Step 1: Data Input

User provides planetary and stellar parameters such as:

- Orbital period
- Planet radius

- Equilibrium temperature
- Insolation flux

Step 2: Feature Processing

- Input is converted into structured format
- Missing features are handled
- Data aligned with trained model schema

Step 3: Model Prediction

- Model processes inputs
- Computes probability using ensemble learning
- Outputs classification

Step 4: Result Display

- Shows:
    - Prediction result
    - Probability score
    - Visual indicators

6. Modules Description

1. Launchpad

- Entry point of application
- Displays key statistics and quick actions

2. Mission Section

- Explains working of exoplanet detection
- Includes:
    - Photometry
    - Feature extraction
    - AI classification

3. Predictor

- Allows user to input parameters
- Provides instant habitability prediction

4. Batch Processing

- Accepts JSON/CSV input
- Processes multiple planets simultaneously

5. Tech Lab

- Shows backend architecture
- API endpoints:
    - /predict
    - /rank

6. Metrics Dashboard

- Displays:
    - Accuracy: 96.8%
    - F1 Score: 0.947
    - AUC-ROC: 0.992
- Includes graphs and confusion matrix

7. Technologies Used

Frontend

- HTML, CSS, JavaScript
- Interactive UI components

Backend

- Python
- Flask

Machine Learning

- Scikit-learn
- Ensemble models

Deployment

- Hosted using Render / Cloud platform

8. Model Performance

- Training Accuracy: 96.8%
- Validation Score: 94.2%
- F1 Score: 0.947
- AUC-ROC: 0.992

The model performs well due to:

- Ensemble learning
- Feature selection
- Noise handling

9. Key Features

- Real-time habitability prediction
- Interactive dashboard
- Batch processing support
- NASA dataset integration
- Feature importance visualization

10. Deployment Details

- Backend deployed using Flask server
- Frontend integrated with API
- Hosted on cloud platform (Render)

11. Future Enhancements

- Add deep learning models
- Real-time NASA API integration
- Improved visualization tools
- Mobile-friendly UI

12. Conclusion

ExoHabitAI successfully demonstrates how machine learning can be used to analyze astrophysical data and identify potentially habitable planets. The system provides a powerful, user-friendly platform for both learning and research purposes.

13. Application Link

Deployed link here:

https://habitability-pzy1.onrender.com